Practice workbook · Midpoints and points that split a line in a ratio

Finding the midpoint of a segment or the point dividing it in a ratio m : n, or working backwards to an end point, a circle centre or a missing vertex.

10 question parts: 1 worked-example retry and 9 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

OLevels.com SCHOOL

Type ID: FPM-COO-02

1. Question 6 · part a · 2 marks

2024 JUN · Paper 01

Worked-example retry · Attempt part a only.

6 The line l passes through the point A with coordinates $(-2, 2)$ and the point B with coordinates $(3, 12)$
The point C with coordinates (p, q) lies on l such that $AC : CB = 3 : 2$

(a) Find the value of p and the value of q

(2)

The line k is perpendicular to l and passes through the point C

(b) Show that an equation of k is $2y + x - 17 = 0$

(4)

The line k crosses the x-axis at the point D

(c) Find the exact length of CD

(3)

The point X with coordinates (m, n) lies on l such that

$$\text{area of triangle } DXC = 80 \text{ units}^2$$

Given that $m > 0$

(d) find the value of m and the value of n

(7)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

P 7 6 5 0 6 A 0 1 2 3 2

1. Question 6 · part a · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

2. Question 11 · part b · 2 marks

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

11 The points A and B have coordinates $(-1, 3)$ and $(5, 6)$ respectively.

(a) Find an equation for the line AB.

(2)

The point P divides AB in the ratio $2 : 1$

(b) Show that the coordinates of P are $(3, 5)$

(2)

The point C with coordinates (m, n), where $m > 0$, is such that CP is perpendicular to the line AB.

Given that the radius of the circle which passes through A, P and C is 5

(c) find the value of m and the value of n.

(6)

The point D with coordinates (p, q) is such that the line AD is perpendicular to the line AB and the line DC is parallel to the line AB.

(d) Find the value of p and the value of q.

(3)

(e) Find the area of trapezium $ABCD$.

(4)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.COM SCHOOL

2. Question 11 · part b · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 9 · part b · 2 marks

2024 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

9 The point A has coordinates $(-4, 3)$ and the point B has coordinates $(6, 8)$
The points A and B lie on the line k

(a) Find an equation of k

(2)

The point C, on k, is such that $AC : CB = 4:1$

(b) Find the coordinates of point C

(2)

The point D with coordinates (p, q), where $p < 0$, lies on the line l through C that is perpendicular to k

The length of CD is $8\sqrt{5}$

(c) Find the coordinates of D

(6)

(d) Find the area of triangle ACD

(2)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

P 7 4 0 9 9 A 0 2 2 3 2

3. Question 9 · part b · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

4. Question 8 · part c · 2 marks

2019 JUN · Paper 02

Further practice · Attempt part c only.

8 The point A has coordinates (2, 6), the point B has coordinates (6, 8) and the point C has coordinates (4, 2).

(a) Find the exact length of

(i) AB (ii) BC (iii) AC

(4)

(b) Find the size of each angle of triangle ABC in degrees.

(3)

The points A, B and C lie on a circle with centre P.

(c) Find the coordinates of P.

(2)

(d) Find the exact length of the radius of the circle in the form $\sqrt{a}$, where a is an integer.

(2)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com — SCHOOL —

4. Question 8 · part c · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

5. Question 9 · part d · 2 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part d only.

9 The line l_1 with equation $y + 2x - 4 = 0$ passes through the point P with coordinates $(a, 6)$ and the point Q with coordinates $(3, b)$.

(a) Find the value of a and the value of b.

(2)

The line l_2 passes through point P and is perpendicular to l_1

The point R, with coordinates (e, f) lies on l_2 such that $PR = 6\sqrt{5}$

(b) Find the two possible pairs of values of e and f.

(8)

Given that $e < 0$,

(c) find the area of triangle PQR.

(3)

The points P, Q and R lie on a circle C.

(d) Find the coordinates of the centre of C.

(2)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

5. Question 9 · part d · 2 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part d only.

Question 9 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

5. Question 9 · part d · 2 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part d only.

Question 9 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

5. Question 9 · part d · 2 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part d only.

Question 9 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

(Total for Question 9 is 15 marks)

5. Question 9 · part d · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

6. Question 4 · part b · 2 marks

2021 JUN · Paper 02

Further practice · Attempt part b only.

4 The point A has coordinates $(-4, -10)$ and the point B has coordinates $(3, 11)$
The line l passes through A and B.

(a) Find an equation of l.

(2)

The point P lies on l such that $AP : PB = 3 : 4$

(b) Find the coordinates of P.

(2)

The point Q with coordinates (m, n), where $m < 0$, lies on the line through P that is perpendicular to l.

Given that the length of PQ is $\sqrt{10}$

(c) find the coordinates of Q.

(6)

The point R has coordinates $(-11, -21)$

(d) Show that

(i) AB and RQ are equal in length,

(ii) AB and RQ are parallel.

(4)

(e) Find the area of the quadrilateral $ABQR$.

(2)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.COM — SCHOOL —

6. Question 4 · part b · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

7. Question 8 · part d · 3 marks

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part d only.

8 The points A and B have coordinates $(1, 5)$ and $(9, 9)$ respectively.

(a) Find an equation of line AB, giving your answer in the form $ax + by + c = 0$, where a, b and c are integers to be found.

(3)

The line l is perpendicular to AB and passes through the point X which lies on AB such that $AX : XB = 3:1$

(b) Show that an equation of l is $y = -2x + 22$

(5)

The point C has coordinates $(6, p)$

Given that C lies on l

(c) find the value of p

(1)

$ABCD$ is a parallelogram where the x coordinate of D is negative.

(d) Find the coordinates of the point D

(3)

(e) Find the area of the parallelogram $ABCD$

(4)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

20

P 7 4 2 8 4 A 0 2 0 3 2

7. Question 8 · part d · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

8. Question 11 · part c · 2 marks

2021 NOV · Paper 02

Further practice · Attempt part c only.

11 The points A and B have coordinates $(-3, -5)$ and $(7, 5)$ respectively.

(a) Find an equation for the line AB

(2)

The point C has coordinates $(p, 1)$ where $p < 0$

Given that AC and BC are perpendicular,

(b) prove that $p = -5$

(7)

The point D, where BCD is a straight line, is such that C divides BD in the ratio $4:3$

(c) Find the coordinates of D

(2)

(d) (i) Find the exact length of AC

(ii) Hence, or otherwise, find the area of triangle ABD

(4)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

8. Question 11 · part c · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

9. Question 8 · part c · 2 marks

2023 JUN · Paper 02

Further practice · Attempt part c only.

8 The points A and B have coordinates $(-6, 8)$ and $(12, 2)$ respectively.

(a) Find an equation of the straight line passing through A and B in the form $ax + by + c = 0$, where a, b and c are integers to be found.

(3)

(b) Find the exact length of AB

(2)

The point X with coordinates (m, n) lies on AB such that $AX:XB = 1:2$

(c) Find the value of m and the value of n

(2)

The line L passes through the point X and is perpendicular to AB
The point C with coordinates (p, q) lies on L where $p > 0$ and $q > 0$

Given that AB is a diameter of a circle and C also lies on the circumference of the circle,

(d) find

(i) the exact value of p

(ii) the exact value of q

(7)

(e) Find the exact area of triangle ABC

(3)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

9. Question 8 · part c · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

10. Question 2 · part c · 2 marks

2023 JAN · Paper 01

Further practice · Attempt part c only.

2 The point A has coordinates $(-5, 3)$, the point B has coordinates $(4, 0)$ and the point C has coordinates $(-1, 5)$.

The line l passes through C and is perpendicular to AB.

(a) Find an equation of l.
Give your answer in the form $ax + by + c = 0$ where a, b and c are integers.

(4)

The line l intersects AB at the point D.

(b) Show that the coordinates of D are $(-2, 2)$.

(3)

(c) Show that l is not the perpendicular bisector of AB.

(2)

(d) Find the value of $\tan \angle ABC$.
Give your answer in its simplest form.

(4)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

10. Question 2 · part c · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL